

JUSTIN TSE

tsejustin0505@gmail.com | linkedin.com/in/tsejustin | (805) 971-9663

EDUCATION

University of California, Berkeley

Master of Landscape Architecture | GPA: 3.77

May 2025

Berkeley, CA

University of California, Santa Barbara

B.A. in Environmental Studies | GPA: 3.62

Jun 2023

Santa Barbara, CA

EXPERIENCE

Research Assistant

Apr 2026 – Present

Institute for the Environment

Hong Kong

- Engineer a **Flask** web application to deliver climate risk and claim settlement insights, including features such as geocoding, user authentication and authorization, spatial aggregation, weather data integration, and data dashboards
- Clean and analyze 10k+ typhoon-induced insurance claim records in **Python** to build geolocations and generate new business insights from complex spatial patterns
- Design verification loops using external geocoding APIs and a **fuzzy string matching** approach to validate geocoding quality
- Collaborate with climate scientists to ideate and execute strategies for applying and communicating **catastrophe modeling** results

Graduate Student Intern (GIS)

Jul 2024 – Dec 2025

City of San Jose

San Jose, CA

- Authored **ETL pipelines** in **FME** to automate the spatial analyses and recurring updates of 800+ parks, 2,700+ amenity assets, multi-year condition assessment records, and census demographics at different spatial scales
- Pioneered a city-wide **multi-criteria decision analysis model** to enhance the prioritization of capital improvement projects
- Implemented **GIS web tools** and **surveys** for field data collection, scenario analysis, project management, and GIS data editing
- Prepared web maps and documentation to document the analytical process and communicate results to diverse stakeholders
- Supported enterprise GIS data management through **versioned editing**, as-built review, **schema design**, and QA/QC operations
- Provided **training and technical support** to cross-functional teams on custom applications, GIS software, and mobile solutions

Graduate Student Researcher

May 2024 – Aug 2025

Institute of Urban and Regional Development

Berkeley, CA

- Implemented end-to-end model training and validation pipelines for hourly PM_{2.5} predictions in **ArcGIS Notebooks** using **low-cost sensor** data and **spatial interpolation principles**
- Benchmarked **machine learning models** against **geostatistical models** to illustrate model performance and latency differences
- Applied **GeoShapley** to measure the joint spatial effects of training features, enhancing machine learning explainability
- Configured an Esri mobile app that synthesizes real-time, daily PM_{2.5} levels and geospatial context for **exposure management**
- Worked with **5+** community organizations to gather functional requirements and conduct **user acceptance testing**

Graduate Student Researcher

May 2024 – Dec 2024

University of California, Berkeley

Berkeley, CA

- Analyzed **700k+** longitudinal hourly sensor measurements to identify patterns and factors related to measurement anomalies
- Created a rule-based Local Climate Zone mapping workflow in **ArcPy** using LiDAR, land cover data, and building polygons
- Developed linear regression-based **calibration models**, enhancing the accuracy of temperature measurements by up to **51%**

PROJECTS

Global Mapping of Local Climate Zone Using the Google DeepMind Satellite Embedding Dataset

2025

- Assisted in developing a **supervised classification** framework to map Local Climate Zone globally using satellite embedding

GIS-based California Tiger Salamander Habitat Suitability and Corridor Analysis

2024

- Performed a pixel-based **habitat suitability analysis** in Suitability Modeler based on human disturbance and hydrological factors
- Streamlined a **regional corridor analysis** using ModelBuilder to identify the least-cost paths between three critical habitats

SKILLS

Languages: Arcade, Python, JavaScript, R, SQL

Frameworks/Libraries: ArcGIS API for Python, ArcPy, Flask, GeoPandas, Leaflet, Rasterio, Sklearn

Computer: Adobe Creative Cloud, ArcGIS Pro, ArcGIS Online, ESRI Suite, FME, Google Earth, Microsoft Office, QGIS, Tableau

CERTIFICATIONS

Graduate Certificate in Geographic Information Science and Technology, University of California, Berkeley

Oracle Cloud Infrastructure Certified Foundation Associate, Oracle

ADDITIONAL TRAINING

NSF CyberTraining Summer School, Texas A&M University

Spatial Data Science: The New Frontier in Analytics, Esri Academy